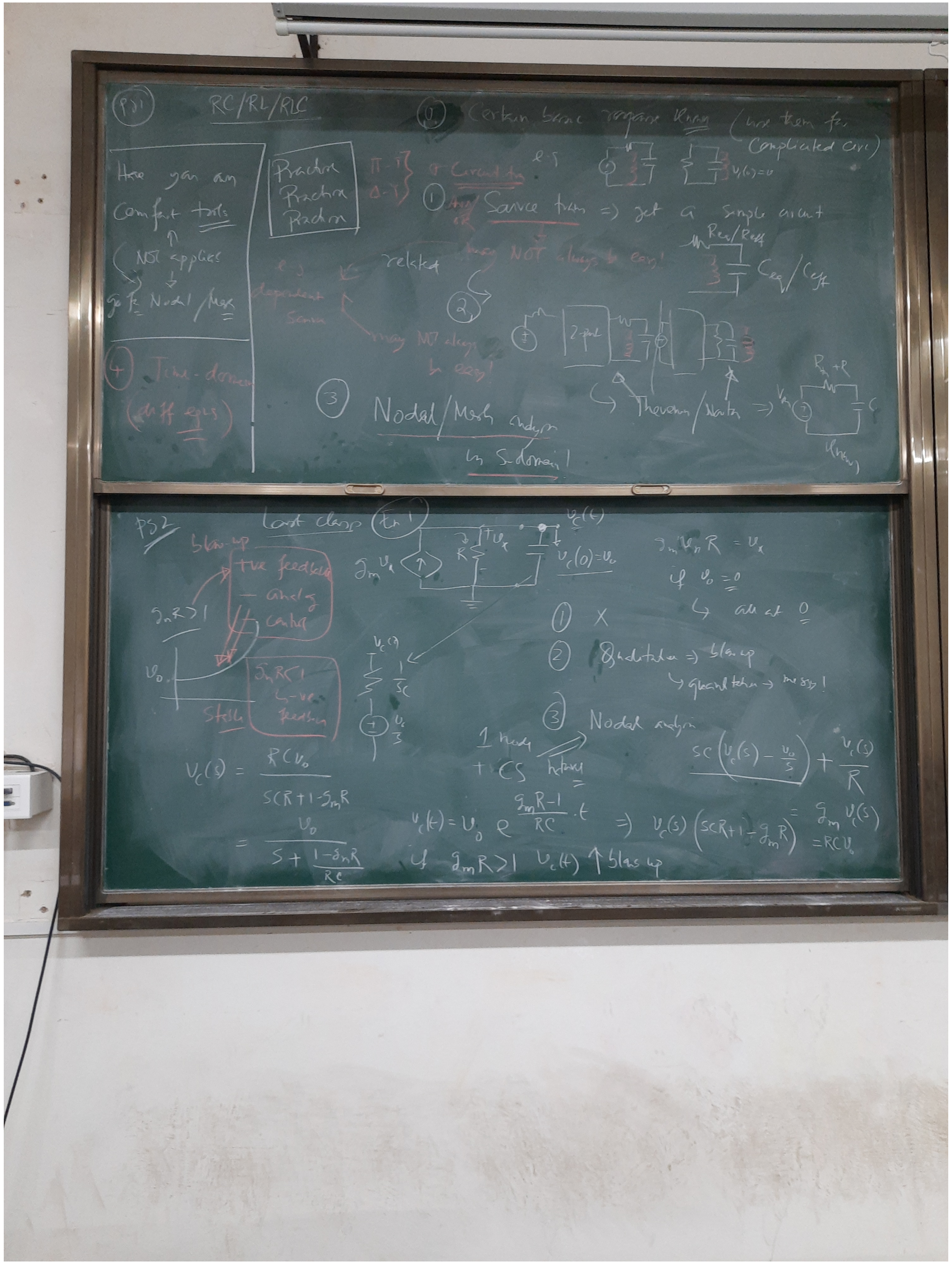


PS1

RC/RL/RLC

Here you are
Comfort tools
↑
NOT applies
↓
go to Nodal / Mesh

④ Time-domain
(diff eqns)

Practice
Practice
Practice

Π-T
Δ-Y } ⓪ Circuit thm

⓪ Certain basic response learning (use them for complicated circ)

e.g. $v_c(0)=0$

① any/or Source trans ⇒ get a simple circuit

may NOT always be easy!

R_{eq}/R_{eff}

C_{eq}/C_{eff}

related

e.g. dependent source

may NOT always be easy!

②

2-port

Thevenin / Norton ⇒ V_{th} $R_{th}+R$ C

$V_{th}(s)$

③ Nodal/Mesh analysis in S-domain!

PS2

Last class (Ex 1)

$g_m v_x$ R $+v_x$ − $v_c(t)$

$v_c(0)=v_0$

$g_m v_x R = v_x$

if $v_0 = 0$
↳ all at 0

① X

② Substitution ⇒ blow up
↳ quantitative → messy!

③ Nodal analysis

1 node
+ CS network

$V_c(s)$ $\frac{1}{sC}$ $\frac{v_0}{s}$

$$sC\left(V_c(s) - \frac{v_0}{s}\right) + \frac{V_c(s)}{R} = g_m V_c(s)$$

$$\Rightarrow V_c(s)\left(sCR + 1 - g_m R\right) = RCv_0$$

$$V_c(s) = \frac{RCv_0}{sCR+1-g_mR} = \frac{v_0}{s + \frac{1-g_mR}{RC}}$$

$$v_c(t) = v_0\, e^{\frac{g_mR-1}{RC}\cdot t}$$

if $g_m R > 1$ $v_c(t)$ ↑ blows up

blow-up

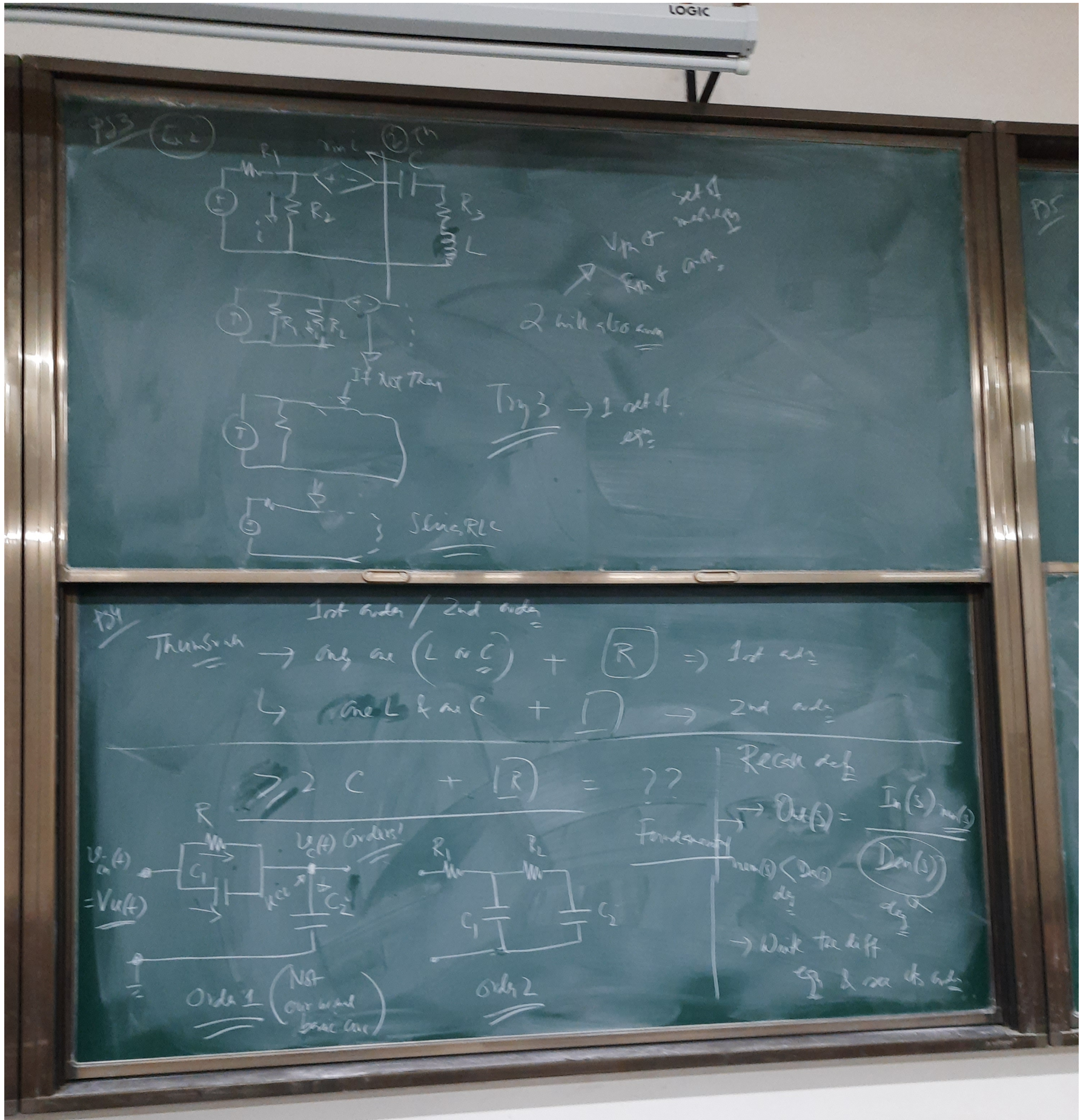
+ve feedback
— analog control

$g_m R > 1$

v_0

Stable

$g_m R < 1$
↳ -ve feedback



Vth & set of mesh eqn
Req & curr.

2 will also come

Try 3 → 1 set of eqn

1st order / 2nd order

Thumb rule → Only one (L or C) + R ⇒ 1st order

↳ one L & one C + R → 2nd order

≥ 2 C + R = ??

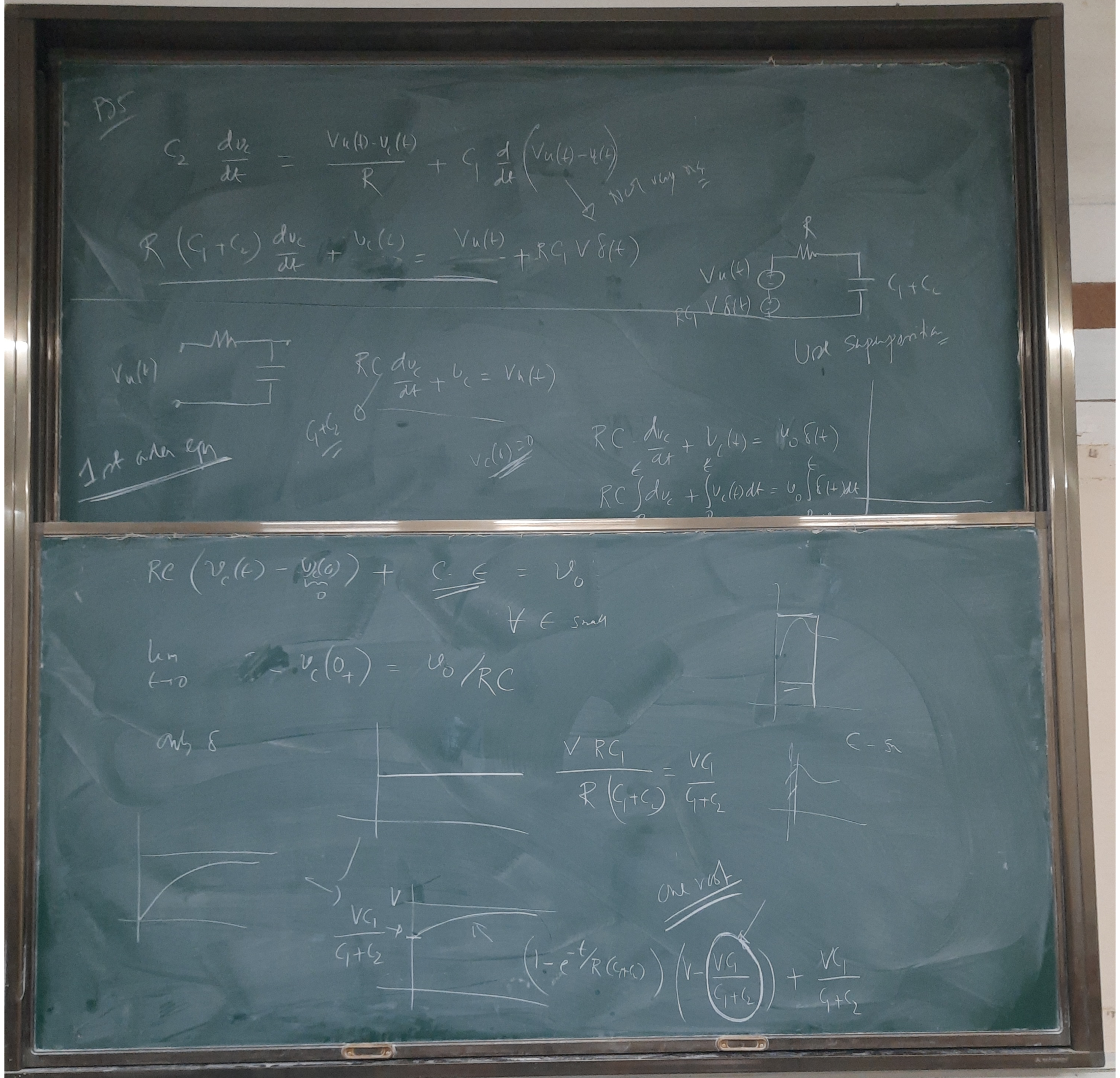
Fundamental

Recall def

$\rightarrow Out(s) = \dfrac{In(s)\, num(s)}{Den(s)}$

num(s) < Den(s)
deg deg

→ Write the diff eqn & see its order.



PS5

$$C_2 \frac{dv_c}{dt} = \frac{V_u(t)-v_c(t)}{R} + C_1 \frac{d}{dt}\left(V_u(t)-v_c(t)\right)$$

NOT very nice

$$R\left(C_1+C_2\right)\frac{dv_c}{dt} + v_c(t) = V_u(t) + RC_1 V\delta(t)$$

R
$V_u(t)$
$RC_1 V\delta(t)$
C_1+C_2

Use superposition

$V_u(t)$

$$RC\frac{dv_c}{dt} + v_c = V_u(t)$$

C_1+C_2

1st order eqn

$v_c(t)=0$

$$RC\frac{dv_c}{dt} + v_c(t) = v_0\delta(t)$$

$$RC\int_0^\epsilon dv_c + \int_0^\epsilon v_c(t)\,dt = v_0\int_0^\epsilon \delta(t)\,dt$$

$$RC\left(v_c(\epsilon) - \underbrace{v_c(0)}_{0}\right) + C\cdot\epsilon = v_0$$

∀ ε small

$$\lim_{\epsilon\to 0} v_c(0_+) = v_0/RC$$

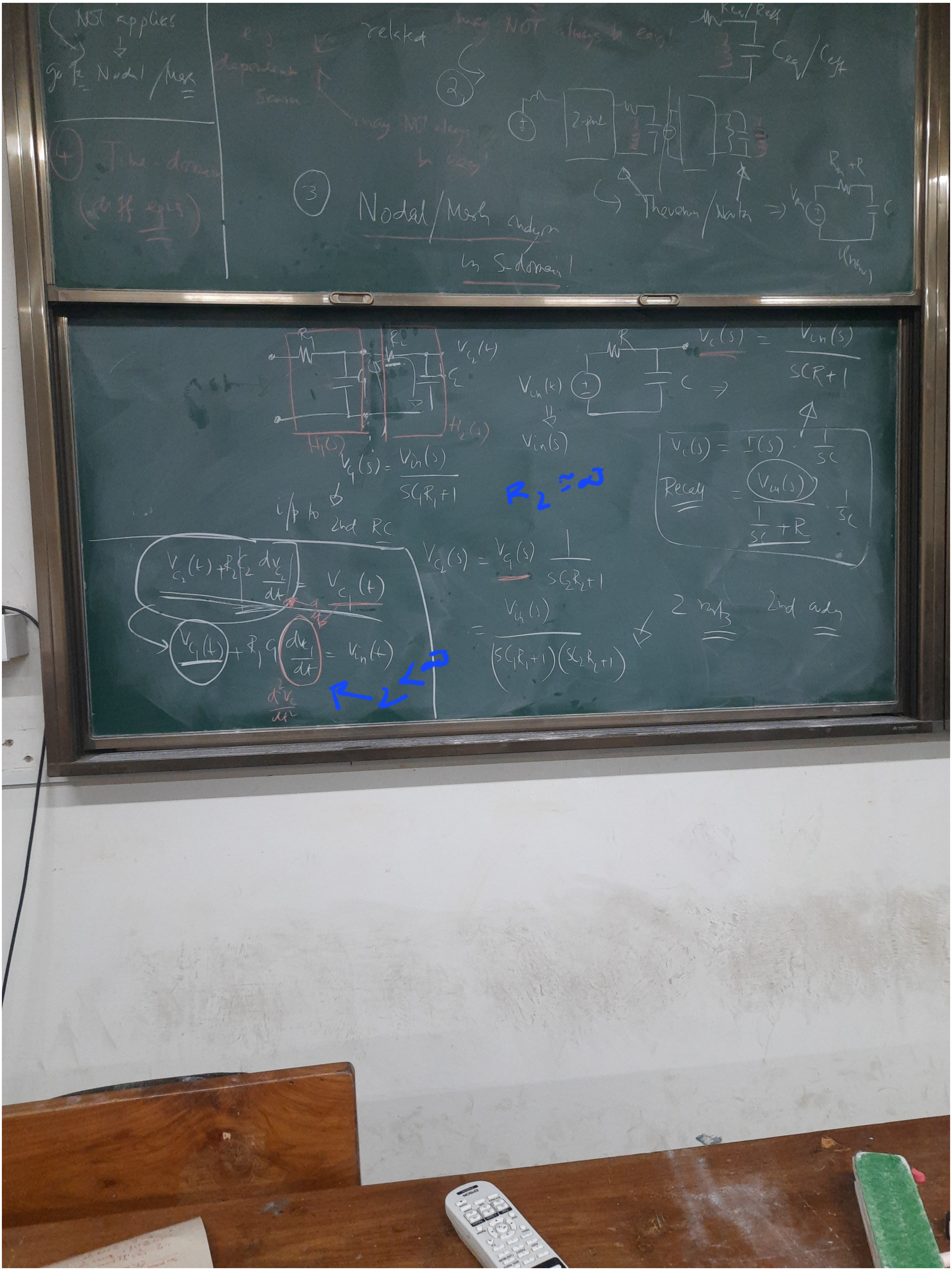
only δ

$$\frac{V\,RC_1}{R\left(C_1+C_2\right)} = \frac{VC_1}{C_1+C_2}$$

ε - sm

one volt

$$\left(1-e^{-t/R(C_1+C_2)}\right)\left(V-\boxed{\frac{VC_1}{C_1+C_2}}\right) + \frac{VC_1}{C_1+C_2}$$



KVL applies
↓
Go to Nodal / Mesh

④ Time-domain (diff eqns)

e.g. dependent sources

related

may NOT always be easy!

②

may NOT always be easy!

③ Nodal/Mesh analysis in S-domain!

$V_{c_1}(s) = \frac{V_{in}(s)}{sC_1R_1+1}$
⇓
i/p to 2nd RC

$R_2 \approx \infty$

$V_{c_2}(t) + R_2C_2\frac{dV_{c_2}}{dt} = V_{c_1}(t)$

$V_{c_1}(t) + R_1C_1\frac{dV_{c_1}}{dt} = V_{in}(t)$

$\frac{d^2V_c}{dt^2}$

$R_2 < \infty$

$V_{c_2}(s) = V_{c_1}(s)\,\frac{1}{sC_2R_2+1}$

$= \frac{V_{in}(s)}{(sC_1R_1+1)(sC_2R_2+1)}$ ⇒ 2 roots — 2nd order

$V_{in}(t)$ ⇓ $V_{in}(s)$

$V_c(s) = \frac{V_{in}(s)}{sCR+1}$ ⇐

$V_c(s) = I(s)\cdot\frac{1}{sC}$

Recall $= \frac{V_{in}(s)}{\frac{1}{sC}+R}\cdot\frac{1}{sC}$